

3. For the n' -th name n and the a' -th attribute a , assign its value $v^*(n, a) = (v_1, \dots, v_C) \in (\mathcal{D}_a)^C$ by setting each v_i as the s_i -th element in $\mathcal{D}_a$, where the integer sequence $(s_1, \dots, s_C) := Q_{N+KD+(n'-1)K+a'} \in [D^C]$.

It is easy to verify that this gives the same dataset distribution as Definition 2.2. Next, consider Q being fixed (thus the dataset $\mathcal{Z}$ being fixed), we construct $P_1, P_2, \dots, P_{N+KD+NK}$ using the given model functions $F^\top(W(\mathcal{Z}), R)$ and $F^\perp(W(\mathcal{Z}), n, a, R)$.

Name part. For the name part, construct P_i for $i \in [N]$ following the approach from the “value-only” warmup example. Specifically, let R_i be fresh randomness, and define $P_i = s$ if $F^\top(W(\{Z\}), R_i)$ matches the s -th element in $\mathcal{N}_0 \setminus \{n_1, \dots, n_{i-1}\}$, or an arbitrary symbol $\emptyset$ if it falls within $\{n_1, \dots, n_{i-1}\}$.²⁹ Adopting the analysis from the “name-only” warmup example, we obtain

$$\sum_{i \in [N]} \log \Pr_{P_i}[P_i = Q_i] = -N \text{loss}_{\text{name}}(\mathcal{Z}) . \quad (\text{F.2})$$

Diversity Part. For the diversity component, we construct the P_i ’s as follows. For each $a' \in [K]$, let a denote the a' -th attribute in $\mathcal{A}$. We form $P_{N+(a'-1)D+i}$ by initially calculating $F^\perp(W(\mathcal{Z}), n, a, R_i)$, where $n \in \mathcal{N}$ is selected uniformly at random.³⁰ Subsequently, if $F^\perp(W(\mathcal{Z}), n, a, R_i)$ corresponds to the s -th element in $\mathcal{T}^L \setminus \{w_1, \dots, w_{i-1}\}$, then set $P_{N+(a'-1)D+i} = s$; otherwise, set $P_{N+(a'-1)D+i} = \emptyset$.³¹

Now, let a be the a' -th element in $\mathcal{A}$. Consider Q as fixed, with randomness arising solely from the calculation of P_i ’s. Note that Q establishes an order of elements in $\mathcal{D}_a$, denoted by $w_1, \dots, w_D$. We have

$$\begin{aligned} \sum_{i \in [D]} \log \Pr_{P_{N+(a'-1)D+i}}[P_{N+(a'-1)D+i} = Q_{N+(a'-1)D+i}] &= \sum_{i \in [D]} \log \mathbb{E}_{n \in \mathcal{N}} \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w_i] \\ &= \sum_{w \in \mathcal{D}_a} \log \mathbb{E}_{n \in \mathcal{N}} \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w] =: \spadesuit_a \end{aligned}$$

Let us denote by $\mathcal{N}_{w,a}$ the set of $n \in \mathcal{N}$ so that $v^*(n, a) = w$. We have

$$\begin{aligned} \spadesuit_a &= \sum_{w \in \mathcal{D}_a} \log \sum_{n \in \mathcal{N}} \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w] - D \log N \\ &\stackrel{\textcircled{1}}{\geq} \sum_{w \in \mathcal{D}_a} \log \sum_{n \in \mathcal{N}_{w,a}} \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w] - D \log N \\ &= \sum_{w \in \mathcal{D}_a} \log \frac{1}{|\mathcal{N}_{w,a}|} \sum_{n \in \mathcal{N}_{w,a}} \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w] - D \log N + \sum_{w \in \mathcal{D}} \log |\mathcal{N}_{w,a}| \\ &\stackrel{\textcircled{2}}{\geq} \sum_{w \in \mathcal{D}_a} \frac{1}{|\mathcal{N}_{w,a}|} \sum_{n \in \mathcal{N}_{w,a}} \log \Pr_R[F_1^\perp(W(\mathcal{Z}), n, a, R) = w] - D \log N + \sum_{w \in \mathcal{D}} \log |\mathcal{N}_{w,a}| \end{aligned}$$

Above, $\textcircled{1}$ uses monotonicity of the log function and $\textcircled{2}$ uses convexity of the log function. Using simple Chernoff bound, one can see that as long as $N \geq \Omega(D \log N)$, with high probability $|\mathcal{N}_{w,a}| \geq$

²⁹Importantly, P_i may depend on $n_1, \dots, n_{i-1}$; however, since Lemma F.1 permits P_i to depend on $Q_1, \dots, Q_{i-1}$, this is acceptable.

³⁰Importantly, $P_{N+(a'-1)D+i}$ depends on $\mathcal{N}$; however, since Lemma F.1 permits P_i to depend on $Q_1, \dots, Q_{i-1}$, and since $\mathcal{N}$ is uniquely determined by $Q_1, \dots, Q_N$, this is acceptable.

³¹Importantly, $P_{N+(a'-1)D+i}$ depends on $w_1, \dots, w_{i-1}$; however, since Lemma F.1 permits P_i to depend on $Q_{N+(a'-1)D+1}, \dots, Q_{N+(a'-1)D+i-1}$, this is acceptable.

$(1 - o(1))\frac{N}{D}$ for all $w \in D$. Thus, we know with high probability

$$\begin{aligned}\spadesuit_a &\geq (1 + o(1))D \sum_{w \in \mathcal{D}_a} \frac{1}{N} \sum_{n \in \mathcal{N}_{w,a}} \log \Pr_R [F_1^\perp(W(\mathcal{Z}), n, a, R) = w] - D \log D - o(D) \\ &= (1 + o(1))D \frac{1}{N} \sum_{n \in \mathcal{N}} \log \Pr_R [F_1^\perp(W(\mathcal{Z}), n, a, R) = v^*(n, a)] - D \log D - o(D)\end{aligned}$$

Thus, summing up over all the diversity part, we have (recall we are fixing Q and thus fixing $\mathcal{Z}$)

$$\begin{aligned}&\sum_{i \in [KD]} \log_{\Pr_{P_{N+i}}} [P_{N+i} = Q_{N+i}] \\ &\geq (1 + o(1))D \frac{1}{NK} \sum_{n \in \mathcal{N}, a \in \mathcal{A}} \log \Pr_R [F_1^\perp(W(\mathcal{Z}), n, a, R) = v^*(n, a)] - KD \log D - o(KD) \\ &= -(1 + o(1))D \text{loss}_{\text{value}1}(\mathcal{Z}) - KD \log D - o(KD) .\end{aligned}\tag{F.3}$$

Value part. For the value part, we construct $P_{N+KD+1}, \dots, P_{N+KD+NK}$ as follows. For $P_{N+KD+(n'-1)K+a'}$, letting n be the n' -th name in $\mathcal{N}$ and a be the a' -th attribute in $\mathcal{A}$. Let us compute $F^\perp(W(\mathcal{Z}), n, a, R)$ and find the corresponding $s_1, \dots, s_C \in [D]$ such that $F_i^\perp(W(\mathcal{Z}), n, a, R)$ is the s_i -th element in $\mathcal{D}_a$ for each $i \in [C]$. If not found, we define $P_{N+KD+(n'-1)K+a'} = \emptyset$; otherwise, define $P_{N+KD+(n'-1)K+a'} = (s_1, \dots, s_C) \in [D^C]$.³²

Following the same simple argument as the “value-only” warmup example, we have

$$\begin{aligned}\sum_{i \in [NK]} \log_{\Pr_{P_{N+KD+i}}} [P_{N+KD+i} = Q_{N+KD+i}] &= \sum_{n \in \mathcal{N}, a \in \mathcal{A}} \Pr [F^\perp(W(\mathcal{Z}), n, a, R) = v^*(n, a)] \\ &= -NK \text{loss}_{\text{value}}(\mathcal{Z})\end{aligned}\tag{F.4}$$

Summing (F.2) (F.3) and (F.4), and applying Lemma F.1, we have

$$\log |\mathcal{W}| \geq \mathbb{E}_{\mathcal{Z}} \left[N \log \frac{N_0 - N}{e^{\text{loss}_{\text{name}}(\mathcal{Z})}} + NK \log \frac{D^C}{e^{\text{loss}_{\text{value}}(\mathcal{Z})}} + KD \log \frac{T^L - D}{De^{(1+o(1))\text{loss}_{\text{value}1}(\mathcal{Z})}} - o(KD) \right] .$$

This finishes the proof of Theorem 3.2. $\square$

G Missing Remark

Remark G.1. Due to the significant overlap among textbooks, especially those designed for PreK-12 education, estimating the total amount of knowledge contained within all English-language textbooks can be challenging. However, we attempt to do so as follows.

According to a 2023 article, Pearson Education, a UK-based educational publisher, reported the highest revenue in 2021, with Wiley and McGraw Hill being the top two US-based educational publishers in terms of revenue.³³

- Pearson’s official website lists fewer than 2,100 textbooks.³⁴

³²Again, importantly, we can do so because $P_{N+KD+(n'-1)K+a'}$ depends on $\mathcal{N}, \mathcal{D}_a$ but they can be computed using the values of $Q_1, \dots, Q_{N+KD}$.

³³<https://wordstrated.com/education-book-publishing-companies-statistics/>, accessed March 2024.

³⁴<https://www.pearson.com/en-us/pearsonplus/search.html> for their full list of eTextbooks and <http://www.mypearsonstore.com/bookstore/browse.asp> for their full list of hard copy books, both accessed March 2024.

- Wiley’s official website lists fewer than 69,000 textbooks.³⁵
- McGraw Hill lists fewer than 22,000 textbooks for PreK-12 education, many of which have significant content overlap (as many are tailored for one of the 50 US states).³⁶ They list fewer than 2,000 textbooks for higher education.

Taking these figures into account, it seems reasonable to estimate that the content of all English-language textbooks could be condensed into no more than 100,000 textbooks. Assuming an average of 160,000 words per book (e.g., 400 pages with 400 words each), this would amount to a total of 16 billion words.

References

- [1] Ibrahim M Alabdulmohsin, Behnam Neyshabur, and Xiaohua Zhai. Revisiting neural scaling laws in language and vision. *Advances in Neural Information Processing Systems*, 35:22300–22312, 2022.
- [2] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of Language Models: Part 1, Context-Free Grammar. *ArXiv e-prints*, abs/2305.13673, May 2023. Full version available at <http://arxiv.org/abs/2305.13673>.
- [3] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of Language Models: Part 3.1, Knowledge Storage and Extraction. *ArXiv e-prints*, abs/2309.14316, September 2023. Full version available at <http://arxiv.org/abs/2309.14316>.
- [4] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of Language Models: Part 3.2, Knowledge Manipulation. *ArXiv e-prints*, abs/2309.14402, September 2023. Full version available at <http://arxiv.org/abs/2309.14402>.
- [5] Zeyuan Allen-Zhu, Yuanzhi Li, and Yingyu Liang. Learning and generalization in overparameterized neural networks, going beyond two layers. *Advances in neural information processing systems*, 32, 2019.
- [6] Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. A convergence theory for deep learning via over-parameterization. In *International conference on machine learning*, pages 242–252. PMLR, 2019.
- [7] Sid Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, USVSN Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of the ACL Workshop on Challenges & Perspectives in Creating Large Language Models*, 2022. URL <https://arxiv.org/abs/2204.06745>.
- [8] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, et al. Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*, 2023.
- [9] William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *The Journal of Machine Learning Research*, 23(1):5232–5270, 2022.
- [10] Elias Frantar, Saleh Ashkboos, Torsten Hoeffler, and Dan Alistarh. GPTQ: Accurate post-training compression for generative pretrained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- [11] Olga Golovneva, Zeyuan Allen-Zhu, Jason Weston, and Sainbayar Sukhbaatar. Reverse training to nurse the reversal curse. *arXiv preprint arXiv:2403.13799*, 2024.
- [12] Suriya Gunasekar, Yi Zhang, Jyoti Aneja, Caio César Teodoro Mendes, Allie Del Giorno, Sivakanth Gopi, Mojan Javaheripi, Piero Kauffmann, Gustavo de Rosa, Olli Saarikivi, et al. Textbooks are all you need. *arXiv preprint arXiv:2306.11644*, 2023.

³⁵<https://www.wiley.com/en-us/subjects>, accessed March 2024. We wrote a code to sum up all the books in all of their subcategories; our code may double count books, so this is only a safe upper bound. We used this number instead of the “21,000” online books mentioned on <https://www.wiley.com/learn/librarysolutions/online-books-purchase.html>, accessed March 2024.

³⁶https://www.mheducation.com/search.html?searchQuery=&page=1&sortBy=title_desc&order=desc&bu-seg&TYPE=Products&PRODUCT_TYPE_PATH=_Student+Materials, accessed March 2024.

- [13] Tom Henighan, Jared Kaplan, Mor Katz, Mark Chen, Christopher Hesse, Jacob Jackson, Heewoo Jun, Tom B Brown, Prafulla Dhariwal, Scott Gray, et al. Scaling laws for autoregressive generative modeling. *arXiv preprint arXiv:2010.14701*, 2020.
- [14] Danny Hernandez, Jared Kaplan, Tom Henighan, and Sam McCandlish. Scaling laws for transfer. *arXiv preprint arXiv:2102.01293*, 2021.
- [15] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*, 2017.
- [16] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.
- [17] Edward J Hu, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*, 2021.
- [18] Changho Hwang, Wei Cui, Yifan Xiong, Ziyue Yang, Ze Liu, Han Hu, Zilong Wang, Rafael Salas, Jithin Jose, Prabhat Ram, Joe Chau, Peng Cheng, Fan Yang, Mao Yang, and Yongqiang Xiong. Tutel: Adaptive mixture-of-experts at scale. *CoRR*, abs/2206.03382, June 2022. URL <https://arxiv.org/pdf/2206.03382.pdf>.
- [19] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- [20] Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. *arXiv preprint arXiv:1705.03551*, 2017.
- [21] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [22] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- [23] Yanzhi Li and Yingyu Liang. Learning overparameterized neural networks via stochastic gradient descent on structured data. In *Advances in Neural Information Processing Systems*, 2018.
- [24] Yanzhi Li, Sébastien Bubeck, Ronen Eldan, Allie Del Giorno, Suriya Gunasekar, and Yin Tat Lee. Textbooks are all you need ii: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023.
- [25] Niklas Muennighoff, Alexander M Rush, Boaz Barak, Teven Le Scao, Aleksandra Piktus, Nouamane Tazi, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. Scaling data-constrained language models. *arXiv preprint arXiv:2305.16264*, 2023.
- [26] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [27] Jonathan S Rosenfeld. Scaling laws for deep learning. *arXiv preprint arXiv:2108.07686*, 2021.
- [28] Jonathan S Rosenfeld, Amir Rosenfeld, Yonatan Belinkov, and Nir Shavit. A constructive prediction of the generalization error across scales. *arXiv preprint arXiv:1909.12673*, 2019.
- [29] Noam Shazeer. Glu variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.
- [30] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2016.
- [31] Jianlin Su, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2021.
- [32] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient